Business Template

BuyEverything Global Retail

Logo / Image

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1 Business Description

1.1 Business background

BuyEverything Global Retail is a volume omnichannel e-commerce platform distributing consumer goods across both domestic US and international markets. In a highly saturated and fast-paced retail ecosystem, operational success depends heavily on a deep understanding of customer buying behavior, supply chain efficiency, and cross-border commercial dynamics. To remain competitive, optimize pricing structures, and minimize overhead costs, the company aggregates huge volumes of raw transactional data from daily digital sales. Capturing this data across separate regional jurisdictions allows the business to track customer demand and financial viability on a global scale of all it can be done by collecting product sales information and analyzing the one using special tools.

1.2 Problems because of poor data management

Without a structured data architecture, the company faces severe data fragmentation silos across its regional operations. The US database tracks metrics using regional state structures, while the International database utilizes separate country-level identifiers with completely different column schemas and naming formats. Poor data governance creates immediate business challenges: incompatible Metrics: variations in field naming conventions, blind spots in profitability

1.3 Benefits from implementing a Data Warehouse

Implementing a centralized Data Warehouse solves structural errors by transforming messy, denormalized source files into a unified single source of truth.

Examples of analytical questions the Data Warehouse will answer:

- What is the true net profitability of specific categories once discounts, shipping overhead, and regional tax obligations are deducted?
- Which customer segments (e.g., Loyal vs. New) exhibit the highest Customer Lifetime Value (CLV) across different geographic areas?
- Return Pattern Analysis: Correlates high product return frequencies (Returned = Yes) with specific device types, regions, or product subcategories to eliminate defective suppliers

1.4 DATASETS DESCRIPTION

The first dataset **US Domestic Operations** contains information about sales on the US market.

The second dataset **International Operations** contains information about sales on the international market.
Both datasets contain similar attributes (name difference is provided in the table below)

International Operations does not contain information about customer gender and age

Location discrepancy - US contains State attribute and international dataset holds Country name

Product Information:

- **PRODUCT_ID**: The unique stock-keeping unit (SKU) identifier assigned to a retail item.
- **PRODUCT_CATEGORY**: The standardized broad structural master classification of the product category.
- **PRODUCT_SUBCATEGORY** : The targeted nested subcategory group mapping exact inventory details.

Customer Information

- **CUSTOMER_ID**: Unique alphanumeric identification key allocated to an individual customer.
- **CUSTOMER_FIRST_NAME**: The legal first given name of the customer profile.
- **CUSTOMER_LAST_NAME**: The family surname of the customer profile.
- **CUSTOMER_EMAIL**: The verified digital mail communication address belonging to the customer.
- **CUSTOMER_AGE**: The chronological age of the customer recorded at the time of purchase.
- **CUSTOMER_GENDER**: The identified gender profile of the consumer.

- **CUSTOMER_SEGMENT**: The specific loyalty or behavioral tracking cohort tier assigned to the consumer.

Sales Representative Information

- **SALES_REP_ID**: Unique identification code assigned to the managing business sales representative.
- **SALES_REP_FIRST_NAME**: The first name of the assigned corporate sales representative.
- **SALES_REP_LAST_NAME**: The last name of the assigned corporate sales representative.
- **SALES_REP_EMAIL**: The internal corporate electronic mail address of the sales representative.

Orders Information

- **ORDER_ID**: Unique identification number assigned to each specific order or sales transaction.
- **ORDER_DT**: The calendar date indicating when the customer transaction was created.
- **ORDER_STATUS**: The execution status of the transaction fulfillment pipeline.
- **RETURNED** A binary flag field indicating whether the item was sent back by the consumer (Yes/No).
- **PAYMENT_METHOD**: The settlement framework or bank asset type used by the customer.
- **UNIT_PRICE**: The base listing price per single individual unit of the catalog item.
- **QUANTITY**: The total number of items or units purchased within this specific transaction.
- **DISCOUNT_PERCENT**: The percentage deduction rate applied to the original base price of the item.
- **DISCOUNT_AMOUNT**: The absolute currency value subtracted from the order total due to applied promotions.
- **SHIPPING_COST**: The logistics and distribution transportation fee required to ship the order.
- **TAX_AMOUNT**: The fiscal sales tax or local value-added tariff applied to the transaction invoice.
- **ORDER_AMOUNT**: The final absolute monetary invoice total charged to the customer after all modifications.
- **COST_AMOUNT**: The comprehensive calculated fulfillment expenditure required to complete the sale.
- **PROFIT_MARGIN_PERCENT**: The calculated net profitability ratio margin structured as a percentage.
- **PROFIT_AMOUNT**: The net financial business revenue remaining after deducting costs from the order total

Location

- **STATE/COUNTRY**: The primary macro-level geographical region identifier (US State name or global Sovereign Country name).
- **CITY** : The localized urban municipality or distribution terminal destination name within the target region.

Sale channel

- **CHANNEL**: The high-level operational marketing and distribution mechanisms through which a consumer places an order.
- **SUBCHANNEL**: The digital platforms or storefront applications that map back to a parent channel category.

US Domestic Operations	International Operations
ORDER_ID	**ORDER_ID_INT**
ORDER_DT	**ORDER_DT_INT**
ORDER_STATUS	**ORDER_STATUS_INT**
RETURNED	**RETURNED_INT**
CUSTOMER_ID	**CUSTOMER_ID_INT**
CUSTOMER_FIRST_NAME	**CUSTOMER_FIRST_NAME_INT**
CUSTOMER_LAST_NAME	**CUSTOMER_LAST_NAME_INT**
CUSTOMER_EMAIL	**CUSTOMER_EMAIL_INT**
CUSTOMER_AGE	

US Domestic Operations	International Operations
CUSTOMER_GENDER	
CUSTOMER_SEGMENT	CUSTOMER_STATUS
SALES_REP_ID	SALES_REP_ID_INT
SALES_REP_FIRST_NAME	SALES_REP_FIRST_NAME_INT
SALES_REP_LAST_NAME	SALES_REP_LAST_NAME_INT
SALES_REP_EMAIL	SALES_REP_EMAIL_INT_INT
PRODUCT_ID	PRODUCT_ID_INT
PRODUCT_CATEGORY	CATEGORY
PRODUCT_SUBCATEGORY	SUBCATEGORY
UNIT_PRICE	PRICE_INT
QUANTITY	QUANTITY_INT
DISCOUNT_PERCENT	DISCOUNT_PERCENT _INT
DISCOUNT_AMOUNT	DISCOUNT_AMOUNT_INT
SHIPPING_COST	SHIPPING_COST_INT
TAX_AMOUNT	TAX
ORDER_AMOUNT	ORDER_AMOUNT_INT
COST_AMOUNT	COST_AMOUNT_INT
PROFIT_MARGIN_PERCENT	PROFIT_MARGIN_PERCENT_INT
PROFIT_AMOUNT	PROFIT_AMOUNT_INT
	COUNTRY
STATE	
CITY	CITY_INT
PAYMENT_METHOD	PAYMENT
CHANNEL	CHANNEL
SUBCHANNEL	SUBCHANNEL

Sales and Cost measures

1. Core Sales Measures (Revenue & Quantities)

These attributes capture the volume and gross financial inputs derived from customer purchasing transactions.

- **UNIT_PRICE / PRICE:** The base, standard retail listing price per single individual unit of a catalog item before any promotional adjustments or regional taxes.

- **QUANTITY / QUANTITY**: The total integer count of item units purchased within a single transaction row, acting as the primary volume multiplier for total sales calculations.
- **ORDER_AMOUNT / ORDER_AMOUNT**: The final absolute gross monetary revenue invoiced to and paid by the customer. It represents the ultimate top-line sales figure for the transaction, calculated as:

2. Cost and Fulfillment Measures

These metrics capture the structural expenses incurred by the business to procure, ship, and settle the order.

- **COST_AMOUNT / COST_AMOUNT_INT**: The total calculated base operational expenditure required to fulfill the transaction. This encompasses the wholesale Cost of Goods Sold (COGS), supply chain inventory procurement, manufacturing costs, or international landed tariffs.
- **SHIPPING_COST / SHIPPING_COST**: The logistics and freight distribution fee required to securely transport the order from the fulfillment warehouse terminal to the customer's specified delivery location.
- **TAX_AMOUNT / TAX**: The unavoidable fiscal sales tax, value-added tax (VAT), or regional duty tariff applied directly to the transaction invoice based on the customer's geographic location.

3. Promotional Reductions

Measures used to track marketing markdown performance and revenue leaks.

- **DISCOUNT_PERCENT / DISCOUNT**: The percentage rate deduction applied to the original base unit price as part of a promotional campaign or customer segment benefit.
- **DISCOUNT_AMOUNT / DISCOUNT_AMOUNT**: The total absolute currency value deducted from the gross order total due to applied promotions or coupons.

4. Net Profitability Measures (Derived Margin Performance)

The final bottom-line health indicators resulting from subtracting business costs from incoming sales amounts.

- **PROFIT_AMOUNT / PROFIT_AMOUNT**: The net financial business revenue remaining after deducting all product fulfillment costs from the top-line order amount.
- **PROFIT_MARGIN_PERCENT / PROFIT_MARGIN_PERCENT**: The structural efficiency ratio representing net profitability, calculated to show what percentage of the incoming sales revenue is captured as clear profit

The hierarchies

These hierarchies based on the available dataset attributes and the business scenarios that can be supported by the model

- Category > Subcategory > Product
- Channel > Subchannel
- Country > State > City

1.5 GRAIN / DIM / FACT

Step 1

The main business process is a sale.

This process captures each completed customer purchase.

Step 2

Declared Grain: One row per sales transaction order.

The source datasets strictly contain single-item transactions; multi-item orders are not provided. Consequently, the order level serves as the most atomic level of granularity available. Because of this traditional order-level summary metrics (such as ORDER_AMOUNT, SHIPPING_COST, and TAX_AMOUNT) may be included as additive measurements at this grain.

Step 3

Identifying dimensional tables

DIM_PRODUCTS

Column name	Description	Data Type
PRODUCT_ID	Surrogate primary key for product identification	INT
PRODUCT_CATEGORY	The standardized broad structural master classification of the product category.	VARCHAR
PRODUCT_SUBCATEGORY	The targeted nested subcategory group mapping exact inventory details.	VARCHAR

Example with filled data

PRODUCT_ID	PRODUCT_SRC_ID	PRODUCT_CATEGORY	PRODUCT_SUBCATEGORY
2	PROD01695	Electronics	Laptop

DIM_CUSTOMERS

Column name	Description	Data Type
CUSTOMER_ID	Surrogate primary key for customer identification	INT
CUSTOMER_FIRST_NAME	The legal first given name of the customer profile.	VARCHAR
CUSTOMER_LAST_NAME	The family surname of the customer profile.	VARCHAR
CUSTOMER_EMAIL	The verified digital mail communication address belonging to the customer.	VARCHAR
CUSTOMER_AGE	The chronological age of the customer recorded at the time of purchase.	INT
CUSTOMER_GENDER	The identified gender profile of the consumer.	VARCHAR
CUSTOMER_SEGMENT	The specific loyalty or behavioral tracking cohort tier assigned to the consumer.	VARCHAR

Example with filled data

CUSTOMER_ID	CUSTOMER_SRC_ID	CUSTOMER_FIRST_NAME	CUSTOMER_LAST_NAME	CUSTOMER_EMAIL	CUSTOMER_AGE	CUSTOMER_GENDER	CUSTOMER_SEGMENT
3	CINT047528	Fabrice	Simon	fabrice.simon047528@hotmail.com	45	Male	Premium

DIM_EMPLOYEES

Column name	Description	Data Type

SALES_REP_ID	Surrogate primary key for sales representative identification	INT
SALES_REP_FIRST_NAME	The first name of the assigned corporate sales representative.	VARCHAR
SALES_REP_LAST_NAME	The last name of the assigned corporate sales representative.	VARCHAR
SALES_REP_EMAIL	The internal corporate electronic mail address of the sales representative.	VARCHAR

Example with filled data

SALES_REP_ID	SALES_REP_SRC_ID	SALES_REP_FIRST_NAME	SALES_REP_LAST_NAME	SALES_REP_EMAIL
2	RINT032	Michael	Turner	michael.turner@novamart.com

DIM_LOCATIONS

Column name	Description	Data Type
COUNTRY_ID	Surrogate primary key for sales representative identification	INT
COUNTRY	The global Sovereign Country name	VARCHAR
STATE	The primary macro-level geographical region identifier (US State).	VARCHAR
CITY	The localized urban municipality or distribution terminal destination name within the target region.	VARCHAR

Example with filled data

COUNTRY_ID	COUNTRY	STATE	CITY
2	Fance	n/a	Paris

DIM_CHANNELS

Column name	Description	Data Type
CHANNEL_ID	Surrogate primary key for channel identification	INT
CHANNEL	The high-level operational marketing and distribution mechanisms through which a consumer places an order.	VARCHAR
SUBCHANNEL	The digital platforms or storefront applications that map back to a parent channel category.	VARCHAR

Example with filled data

CHANNEL_ID	CHANNEL	SUBCHANNEL
1	Mobile App	iOS App

DIM_DATE

Column name	Description	Data Type
ORDER_DT	The calendar date indicating when the customer transaction was created.	DATE

DIM_ORDER_DETAILS (JUNK DIMENSION OF UNRELATED ATTRIBUTES)

Column name	Description	Data Type
ORDER_ID	Surrogate primary key for order identification	INT
ORDER_STATUS	The execution status of the transaction fulfillment pipeline.	VARCHAR
RETURNED	Binary flag indicating whether the item was returned by the customer (Yes/No).	VARCHAR
PAYMENT_METHOD	The settlement framework or bank asset type used by the customer.	VARCHAR

Example with filled data

ORDER_ID	ORDER_SRC_ID	ORDER_STATUS	RETURNED	PAYMENT_METHOD
2	OINT000001	Delivered	No	Wallet

Step 4

Identifying fact table

FCT_ORDERS_DD

Column name	Description	Data Type
ORDER_ID	Unique identification number assigned to each specific order or sales transaction	VARCHAR
PRODUCT_ID	The unique stock-keeping unit (SKU) identifier assigned to a retail item	VARCHAR
SALES_REP_ID	Unique identification code assigned to the managing business sales representative.	VARCHAR
CUSTOMER_ID	Unique alphanumeric identification key allocated to an individual customer	VARCHAR
QUANTITY	The total number of items or units purchased within this specific transaction	INT
UNIT_PRICE	The base listing price per single individual unit of the catalog item.	NUMERIC(10,2)

DISCOUNT_PERCENT	The percentage deduction rate applied to the original base price of the item.	INT
DISCOUNT_AMOUNT	The absolute currency value subtracted from the order total due to applied promotions.	NUMERIC(10,2)
SHIPPING_COST	The logistics and distribution transportation fee required to ship the order	NUMERIC(10,2)
TAX_AMOUNT	The fiscal sales tax or local value-added tariff applied to the transaction invoice	NUMERIC(10,2)
ORDER_AMOUNT	The final absolute monetary invoice total charged to the customer after all modifications.	NUMERIC(10,2)
COST_AMOUNT	The comprehensive calculated fulfillment expenditure required to complete the sale.	NUMERIC(10,2)
PROFIT_MARGIN_PERCENT	The calculated net profitability ratio margin structured as a percentage.	NUMERIC(10,2)
PROFIT_AMOUNT	The net financial business revenue remaining after deducting costs from the order total	NUMERIC(10,2)

Example with filled data

ORDER_ID	PRODUCT_ID	SALES_REP_ID	CUSTOMER_ID	QUANTITY	UNIT_PRICE	DISCOUNT_PERCENT	DISCOUNT_AMOUNT	SHIPPING_COST	TAX_AMOUNT	ORDER_AMOUNT	COST_AMOUNT	PROFIT_MARGIN_PERCENT	PROFIT_AMOUNT
4	4	7	5	1	12.76	5	18.38	0	22.18	371.46	121.24	12.25	39.1

2 Business Layer 3NF

The entities that has transitive dependencies and were normalized:

- Order > Customer
- Order > Product > Subcategory > Category
- Order > Employee
- Order > City > State > Country
- Order > Subchannel>Channel

● Customer information was selected as the most suitable candidate for SCD Type 2 implementation because customer attributes may change over time while historical values remain important for analysis.
● In situations where source datasets do not provide values for required attributes, placeholder values such as 'N/A' will be loaded during the ETL process. Default constraints were intentionally not used, following the recommendations provided during the course.
● The source datasets do not provide identifiers for product categories and product subcategories. Therefore, PRODUCT_CATEGORY_NAME and PRODUCT_SUBCATEGORY_NAME serve as the business identifiers and source identifiers for those entities.
● The dataset contains one product per order. As a result, a separate order line table was not required, and

product information can be directly associated with each order record.

3 Business Layer Dimensional Model

4 Logical Scheme

5 Data Flow

6 Fact Table Partitioning Strategy